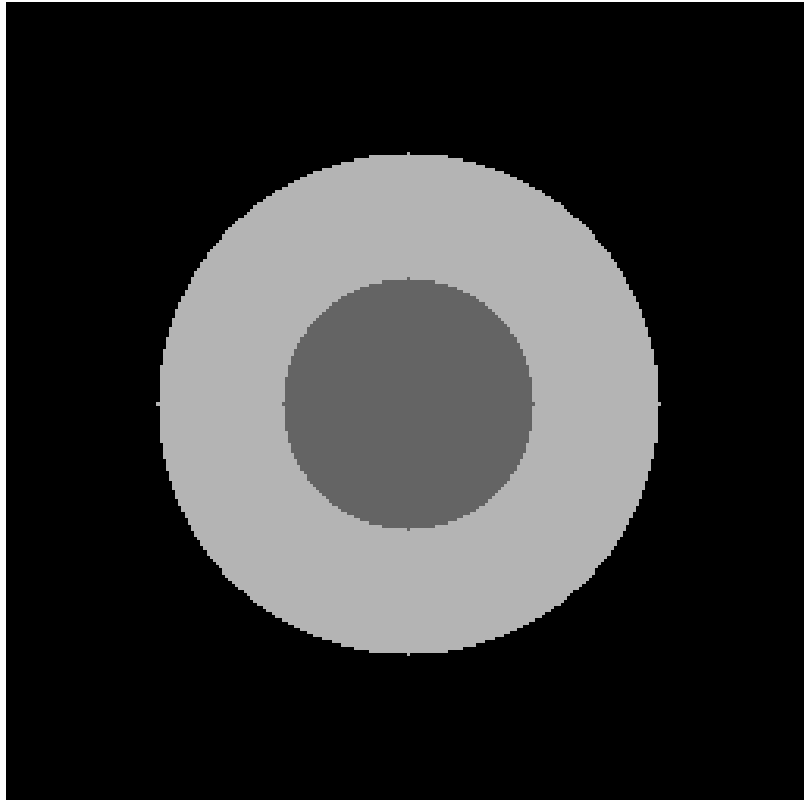


Pixel vs Intensity Resolution in Medical Imaging

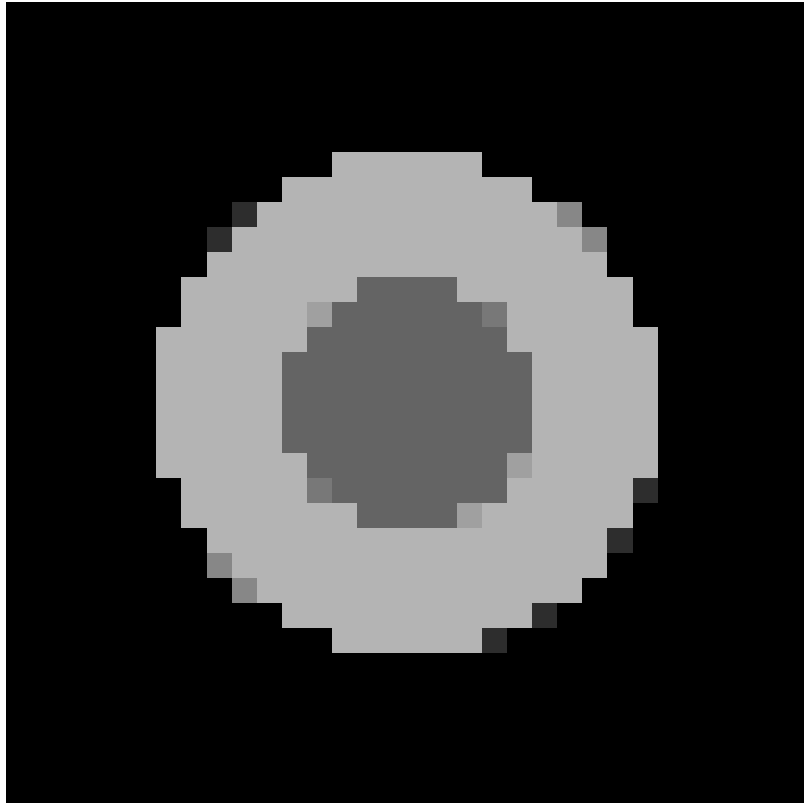
Step 1: Input Image (Original)

This image represents a high-quality grayscale medical-like image. It contains smooth intensity variations and sharp boundaries. It preserves both spatial details (shape clarity) and intensity details (contrast differences), which are essential for accurate diagnosis.



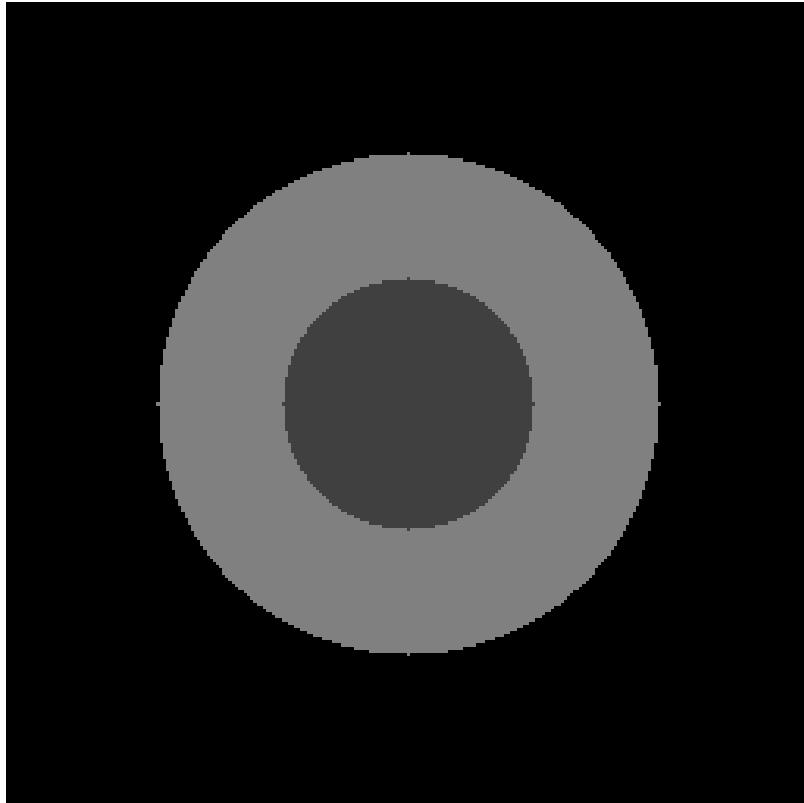
Step 2: Low Pixel Resolution

The image is downsampled, reducing the number of pixels. This leads to a blocky or pixelated appearance. Edges become jagged and fine details are lost. In medical imaging, this affects the ability to detect small structures like tumors or lesions.



Step 3: Low Intensity Resolution

The number of gray levels is reduced, causing a banding effect. Smooth transitions are lost and appear as distinct layers. This reduces contrast resolution, making it difficult to distinguish between tissues with similar intensity values.



Conclusion

Pixel resolution determines spatial detail, while intensity resolution determines contrast detail. In medical imaging, both are critical: high pixel resolution helps detect small structures, and high intensity resolution helps differentiate tissues.